Natural Language Processing

Assignment 04 Solutions

1. **You are building a Maximum Entropy model for a language with a vocabulary size of $V = 1050$. You know that a specific token $\langle UNK \rangle$ has a probability of 0.1. Additionally, a specific set of 49 "closed-class" words shares a total probability mass of 0.4. With no other constraints, what is the probability assigned to the word "algorithm" (which is neither $\langle UNK \rangle$ nor in the closed-class set)? [2 Marks]**

   (a) 5.0×10^{-4}

   (b) 4.76×10^{-4}

   (c) 1.0×10^{-3}

   (d) 4.0×10^{-4}

   Correct Answer: A
   Explanation: The MaxEnt principle dictates modeling known constraints and assuming uniformity for the unknown.
   1. Total Probability Mass = 1.0.
   2. Known Mass = $P(\langle UNK \rangle) + P(\text{Closed}) = 0.1 + 0.4 = 0.5$.
   3. Remaining Mass to distribute = $1.0 - 0.5 = 0.5$.
   4. Count of Known Types = $1(\langle UNK \rangle) + 49(\text{Closed}) = 50$.
   5. Remaining Vocabulary = $1050 - 50 = 1000$ words.
   6. $P(\text{"algorithm"}) = \frac{\text{Remaining Mass}}{\text{Remaining Vocab}} = \frac{0.5}{1000} = 0.0005 = 5.0 \times 10^{-4}$.

2. **Consider a MaxEnt model for POS tagging predicting a tag $y \in \{N, V\}$ given context x.**
   Features: $f_1(x, y) = 1$ if $y = N$ (weight $\lambda_1 = \ln 2$), $f_2(x, y) = 1$ if $y = V$ (weight $\lambda_2 = \ln 3$). Calculate the conditional probability $P(N|x)$. [2 Marks]

   (a) 0.2

   (b) 0.4

   (c) 0.5

   (d) 0.6

   Correct Answer: B
   Explanation:
   Numerator for N: $\exp(\lambda_1 \times 1) = \exp(\ln 2) = 2$.
   Numerator for V: $\exp(\lambda_2 \times 1) = \exp(\ln 3) = 3$.
   Normalization $Z(x) = 2 + 3 = 5$.
   $P(N|x) = 2/5 = 0.4$.

3. **Suppose you have the input sentence "Standard approaches yield poor results". And you know the possible tags each of the words in the sentence can take:**

   - Standard: JJ, NN
   - approaches: NNS, VBZ
   - yield: VB, VBP, NN
   - poor: JJ
   - results: NNS, VBZ

How many possible hidden state sequences are possible for the above sentence and States? [1 Marks]

(a) $2 \times 2 \times 2 \times 2$
(b) 5×3
(c) $2 + 2 + 3 + 1 + 2$
(d) $2 \times 2 \times 3 \times 2$

Correct Answer: D
Explanation: The total number of possible hidden state sequences is the Cartesian product of the possible tags for each word in the sequence.
Possibilities per word:
Standard (2) $\times$ approaches (2) $\times$ yield (3) $\times$ poor (1) $\times$ results (2)
Total $= 2 \times 2 \times 3 \times 1 \times 2 = 24$.
Option D represents this product $(2 \times 2 \times 3 \times 2)$.

4. **Once a day (e.g. at noon), the weather is observed as one of three states: State 1: Rainy (R), State 2: Cloudy (C), State 3: Sunny (S). The state transition probabilities are given in the table below:**

	Rainy (1)	**Cloudy (2)**	**Sunny (3)**
Rainy (1)	0.4	0.3	0.3
Cloudy (2)	0.2	0.6	0.2
Sunny (3)	0.1	0.1	0.8

Given that the weather on day 1 ($t = 1$) is Rainy (State 1), what is the probability that the weather sequence for the next 4 days will be "Cloudy - Cloudy - Sunny - Rainy"? [Marks 2]

(a) 3.6×10^{-3}
(b) 2.4×10^{-2}
(c) 1.4×10^{-3}
(d) 5.0×10^{-4}

Answer: A
Explanation:
We are looking for the probability of the sequence $O = \{S_1, S_2, S_2, S_3, S_1\}$ given the model and starting at S_1.

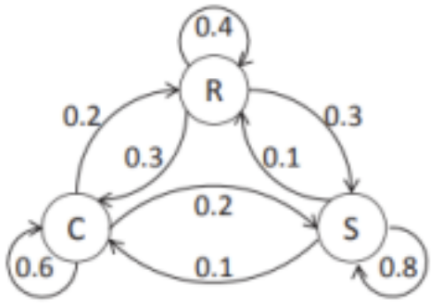
The sequence of states is:

- Day 1: Rainy (S_1) [Given]
- Day 2: Cloudy (S_2)
- Day 3: Cloudy (S_2)
- Day 4: Sunny (S_3)
- Day 5: Rainy (S_1)

$$P(O|Model) = P(S_1) \times P(S_2|S_1) \times P(S_2|S_2) \times P(S_3|S_2) \times P(S_1|S_3)$$

Using the transition matrix values:

$$\begin{aligned} &= (1) \times (0.3) \times (0.6) \times (0.2) \times (0.1) \\ &= 0.3 \times 0.6 \times 0.02 \\ &= 0.18 \times 0.02 \\ &= 0.0036 \\ &= 3.6 \times 10^{-3} \end{aligned}$$



5. **In the context of the weather model above, what is the expected number of consecutive days the weather remains Cloudy?** [Marks 1]

   (a) 1.67
   (b) 5.0
   (c) 2.5
   (d) 10.0

   Answer: C

   Explanation:
   The expected duration (expected number of consecutive days) in a state i is given by the formula for the mean of a geometric distribution:

$$E_i = \frac{1}{1 - a_{ii}}$$

   For the Cloudy state (State 2), the self-transition probability a_{22} is 0.6.

$$E_{Cloudy} = \frac{1}{1 - 0.6} = \frac{1}{0.4} = 2.5$$

6. **A stock market analyst models the market behavior as having two hidden states: Bull Market (B) and Bear Market (BE). We cannot observe the state directly, but we can observe the daily trade volume, which can be High (H), Medium (M), or Low (L). Let us start on Day 1 in a Bull Market state ($q_1 = B$). The HMM parameters are given below:**

Transition Probabilities (a_{ij})

	Bull (B)	**Bear (BE)**
Bull (B)	0.7	0.3
Bear (BE)	0.4	0.6

Emission Probabilities ($b_j(k)$)

	High (H)	**Medium (M)**	**Low (L)**
Bull (B)	0.6	0.3	0.1
Bear (BE)	0.1	0.3	0.6

   What is the probability that the observation on Day 2 is "Low" ($O_2 = L$)? [Marks 2]

   (a) 0.45
   (b) 0.07
   (c) 0.25
   (d) 0.32

Correct Answer: C
Explanation:
We are given $q_1 = B$. We need to find $P(O_2 = L)$. Since we don't know the state on Day 2, we must sum over all possible transitions from Day 1 to Day 2.

$$P(O_2 = L) = P(O_2 = L|q_2 = B)P(q_2 = B|q_1 = B) + P(O_2 = L|q_2 = BE)P(q_2 = BE|q_1 = B)$$

$$= (P(L|B) \times P(B|B)) + (P(L|BE) \times P(BE|B))$$

$$= (0.1 \times 0.7) + (0.6 \times 0.3)$$

$$= 0.07 + 0.18$$

$$= 0.25$$